

02 — Simulation: families, prior, constraints

17 September 2026

1 Theory

Aim

- Generate pool of patterns swept over the regime coordinates $(\bar{n}, \tilde{\delta})$; train and test are partitioned from it later
- Same $(\bar{n}, \tilde{\delta})$ distribution in every structured family, so regimes carry no info about the family
- Matérn II: swept on $\tilde{\delta}$ like the rest, but its regimes are reported in μ_R (hard core in pairs, 01), since $\tilde{\delta}$ misses hard cores below $r_{\min}(\bar{n})$
- Two patterns per θ so we can compute bias/variance at fixed θ

Families

Each $\theta = \bar{n} + \text{shape}$ (drawn) + *amplitude* (solved for the target $\tilde{\delta}$).

Family	Model	Shape	Amplitude
Poisson	homogeneous, intensity $\bar{n}$	—	—
Thomas	parents κ ; Poisson(μ) offspring, $N(0, \sigma^2 I)$; $\bar{n} = \kappa\mu$	σ	μ
Nested	meta-parents κ ; Poisson(μ_1) parents, $N(0, \sigma_1^2 I)$; Poisson(μ_2) children, $N(0, \sigma_2^2 I)$; $\bar{n} = \kappa\mu_1\mu_2$	$\sigma_2, \rho = \sigma_1/\sigma_2, \mu_1$	μ_2
LGCP	$\Lambda = e^{\mu+Y}$, $\text{Cov } Y = \sigma^2 e^{-r/s}$; $\bar{n} = e^{\mu+\sigma^2/2}$	s	σ^2
Matérn II	primaries λ_p , hard core R (type II); $\bar{n} = (1 - e^{-\lambda_p \pi R^2})/(\pi R^2)$	—	R

Shape and amplitude

- Shape = where the structure lives (its length scale); amplitude = how strong it is at that scale
- At fixed $(\bar{n}, \text{shape})$, amplitude $\rightarrow 0$ gives CSR; raising it strengthens the structure without moving it:
 - Thomas μ : same $\bar{n}$ in fewer, bigger clusters ($\kappa = \bar{n}/\mu$); nested μ_2 : same at the inner level
 - LGCP σ^2 : more variable intensity; Matérn R : bigger hard core
- $\tilde{\delta}$ increases monotonically with the amplitude (checked on grids), so each target $\tilde{\delta}$ has exactly one amplitude; solved by root-finding on the closed-form K (`families.py`)

Validity rules

Each rule removes θ for which $\tilde{\delta}$ would be misleading. Rules 1–2 restrict the shape (a box of allowed shapes at each $\bar{n}$); rules 3–4 cap the amplitude. Two tolerances: $\varepsilon = 0.2$, $c = 0.45$. Radius set $R(\bar{n}) = [r_{\min}(\bar{n}), 0.25]$ as in 01.

1. **Scale inside** $R(\bar{n})$ (shape)

- Why: $\tilde{\delta}$ only looks at $R(\bar{n})$. Structure mostly below $r_{\min}(\bar{n})$ or beyond 0.25 is missed, so patterns with the same $\tilde{\delta}$ would not be equally hard
 - Rule: $\gamma(r) \in [0, 1]$ = how much of the structure is still to come at r (Thomas: $e^{-r^2/4\sigma^2}$, fraction of cluster pairs farther apart than r ; LGCP: $e^{-r/s}$). Need $\gamma(r_{\min}(\bar{n})) \geq \varepsilon$ and $\gamma(0.25) \leq \varepsilon$
 - Gives: $\sigma \in [r_{\min}(\bar{n}), 0.25] / 2\sqrt{\ln(1/\varepsilon)}$ ($\sigma \leq 0.098$); $s \in [r_{\min}(\bar{n}), 0.25] / \ln(1/\varepsilon)$ ($s \leq 0.155$); nested: inner σ_2 for the lower bound, outer $\sqrt{\sigma_1^2 + \sigma_2^2}$ for the upper
2. **Nested levels distinguishable** (shape)
- Why: otherwise nested is just a Thomas process
 - Separated: inner level done ($\gamma_{\text{in}} \leq \varepsilon$) before the outer one starts ($\gamma_{\text{out}} \geq 1 - \varepsilon$) $\Rightarrow 1 + \rho^2 \geq \ln(1/\varepsilon)/(-\ln(1 - \varepsilon))$, i.e. $\rho \geq 2.49$
 - Balanced: inner level carries at least ε of the K excess, $1/(1 + \mu_1) \geq \varepsilon \Rightarrow \mu_1 \leq 4$; also $\mu_1 \geq 1$ (interpretability)
3. **Enough independent units** (amplitude)
- Why: $\tilde{\delta}$ is computed at $\bar{n}$, so it only means something if $n \approx \bar{n}$. Few big clusters make n swing
 - Rule: $\text{CV}(n) = \text{sd}(n)/\bar{n} \leq c$, with

$$\text{CV}^2 = \frac{1}{\bar{n}} + \int_0^1 \bar{\gamma}_W(r) d(K(r) - \pi r^2), \quad \bar{\gamma}_W(r) = 1 - \frac{4r}{\pi} + \frac{r^2}{\pi} \text{ (isotropised set covariogram of } W\text{)}$$

- Formula vs simulated Thomas CV: 0.409/0.413, 0.310/0.306, 0.444/0.445
 - Old DV3 in these terms: Thomas 0.41–0.44, nested 0.26–0.31, LGCP up to 0.72 (inconsistent)
4. **Matérn II packing** (amplitude)
- Why: type II thinning cannot pass fill $\pi R^2 \bar{n} = 1$; near it the process degenerates
 - Rule: $\pi R^2 \bar{n} \leq 0.95$ ($\lambda_p \leq 3.15 \bar{n}$)
- With $\varepsilon = 0.2$ we recover the old DV3 bounds ($\sigma \leq 0.1$, $s \leq 0.15$, $\rho \geq 3$); only the LGCP lower bound on s is wider ($0.5 \rightarrow 0.16$ – 0.055 spacings).

Feasibility and the $\tilde{\delta}$ range

Largest $\tilde{\delta}$ reachable under the rules (`scripts/feasibility.py`):

$\bar{n}$	100	152	230	348	528	800
Thomas	16.8	26.9	42.3	65.5	101	155
Nested	14.4	23.9	38.7	61.7	97.1	150
LGCP	17.3	27.4	42.3	63.9	95.6	148
Matérn II	4.3	5.3	6.5	8.0	9.9	11.9

- Cluster families $\propto \bar{n}$ (CV fixes the clustering, $s_0 \propto 1/n$); Matérn $\propto \sqrt{\bar{n}}$ (packing limit)
- Matérn cap physical: fill $0.95 \rightarrow 0.999$ moves it $4.34 \rightarrow 4.45$ at $\bar{n} = 100$
- Common range at every $\bar{n} \Rightarrow \tilde{\delta} \in [\delta_{lo}, 4]$; $\bar{n}$ and $\tilde{\delta}$ independent
- $\delta_{lo} = 0.1$ for now; any larger δ_{lo} is a subset of this pool (log-uniform restricts to log-uniform)

Prior (draw order is part of the definition)

1. $\bar{n} \sim \text{log-uniform}[100, 800]$
 2. $\tilde{\delta} \sim \text{log-uniform}[0.1, 4]$ (Poisson: $\tilde{\delta} = 0$, skip 2–4)
 3. Shape: each coordinate log-uniform on its rule-1/2 box at $\bar{n}$; redraw until valid, $\tilde{\delta}$ reachable, and (LGCP) a grid exists
 4. Amplitude: the unique a with $\tilde{\delta}(\bar{n}, \text{shape}, a) = \text{target}$ (root in $\log a$), must be below the rule-3/4 cap
- Amplitude is not drawn, so its distribution is whatever the $\tilde{\delta}$ prior induces (Fig. 4)

Samplers

- Poisson: $N \sim \text{Poisson}(\bar{n})$ uniform points
- Thomas / nested: top-level parents on W expanded by $5\times$ the total displacement s.d. (missed mass $e^{-12.5}$), offspring, crop to W
- Matérn II: primaries on $W \oplus R$ with $U(0,1)$ marks, delete a point if a neighbour within R has a smaller mark, crop (exact)
- LGCP: circulant embedding on the $(2M)^2$ torus, spacing $1/M$: $Y = \mathcal{F}^{-1}[\sqrt{\lambda}\mathcal{F}w]$, w white noise, $\lambda = \mathcal{F}(\text{covariance})$ (exact on the grid); Poisson cell counts, uniform in cells
 - M : smallest power of two ≥ 128 with $|K_{\text{grid}} - K| \leq 0.1 \cdot 2\pi r s_0(r; \bar{n})$ on $R(\bar{n})$
 - Grid K : bilinear interpolation of e^C on the lattice, midpoint rule at $1/(4M)$
- Patterns conditioned on $n \in [20, 2000]$ (tables' domain), redrawn from the same stream

2 Implemented pipeline

Code

src/cloudforger/simulation/seeding.py	addressed random streams
src/cloudforger/simulation/families.py	rules, shape boxes, $K - \pi r^2$, CV, $\tilde{\delta}$, amplitude solve
src/cloudforger/simulation/processes.py	the five samplers
src/cloudforger/simulation/lgcp_grid.py	LGCP grid size
src/cloudforger/simulation/sweep.py	θ draw, replicate patterns, shards
src/cloudforger/simulation/check.py	mean $\hat{K}$ vs closed form
scripts/simulate.py	CLI: run , merge , check
scripts/feasibility.py	maximum reachable $\tilde{\delta}$
scripts/simulation_figures.py	figures of this document
configs/simulation/config.yaml	$\bar{n}$, $\tilde{\delta}$ range, rules, sizes

Configuration

- 10 000 θ per family $\times$ 2 patterns = 20 000 patterns per family, 100 000 total
- $\bar{n} \in [100, 800]$, $\tilde{\delta} \in [0.1, 4]$, $\varepsilon = 0.2$, $c = 0.45$, fill 0.95, $\mu_1 \geq 1$; shards of 250 θ

Seeding

- Key (DV = 3, set, family, index, role), root shared with 01
- θ : (**sweep**, family, i , PARAMS); pattern: (**sweep**, family, $2i + \text{rep}$, PATTERN)
- Checks: (**pilot**, family, $10^9 + 10^4 i + p$); gallery (**pilot**, $2 \cdot 10^9 + \dots$); both disjoint from the null curves of 01
- More θ later: append indices, nothing existing changes

Steps

1. **Run** (**run --jobs 10**; 3 min 10 s on 10 cores): one file per shard, atomic, resumable
2. **Merge** (**merge**): per family **points.npz** (points, offsets) + **manifest.csv** in **data/simulation/**; ~ 106 MB each
 - Manifest: **case_id** (**family-theta-rep**), $\bar{n}$, $\tilde{\delta}$, CV, amplitude, shape, model parameters, LGCP M , n , tries
3. **Check** (**check**): 8 θ per family, 500 patterns each; $\tilde{K} = \hat{K} n(n-1)/\bar{n}^2$ (unbiased) vs closed form on 16 radii of $R(\bar{n})$

Results

Check	Result	Reference
Closed-form K vs old DV3 implementation	$\leq 10^{-10}$	—
Sampler $\tilde{K}$ vs K , max $ z $	1.6–2.8 by family	$\lesssim 3.4$ (16 radii)
Sampler $\tilde{K}$ vs K , max rel. error	$\leq 3.2\%$; Matérn 10% at $K \approx 0$	—
$\log \tilde{\delta}$ uniform, KS distance	≤ 0.0075	0.0136 (1%)
Shape draws per θ	1; nested mean 1.96, max 14	—
Pattern redraws ($n \notin [20, 2000]$)	0; realised n 44–961	—
Realised max CV	Thomas 0.32, nested 0.30, LGCP 0.34	$c = 0.45$
LGCP grid M (128/256/512/1k/2k/4k)	8064/1013/566/268/76/13 θ	—
5% L-test size, Poisson pool	0.049 (20 000 patterns)	0.05

3 Regimes

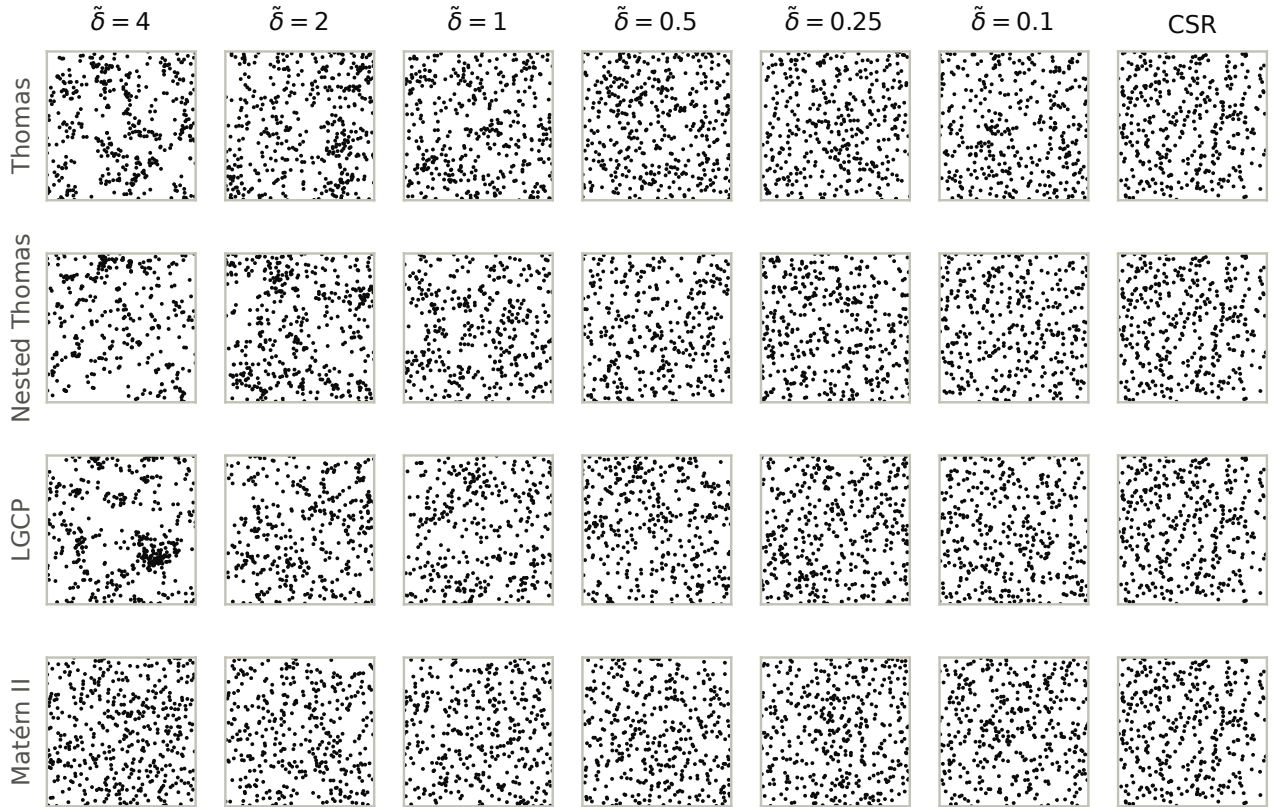


Figure 1: One pattern per cell, $\bar{n} = 400$, fixed shape per family (Thomas $\sigma = 0.03$; nested $\sigma_2 = 0.008$, $\rho = 5$, $\mu_1 = 3$; LGCP $s = 0.04$). Only the amplitude changes along a row.

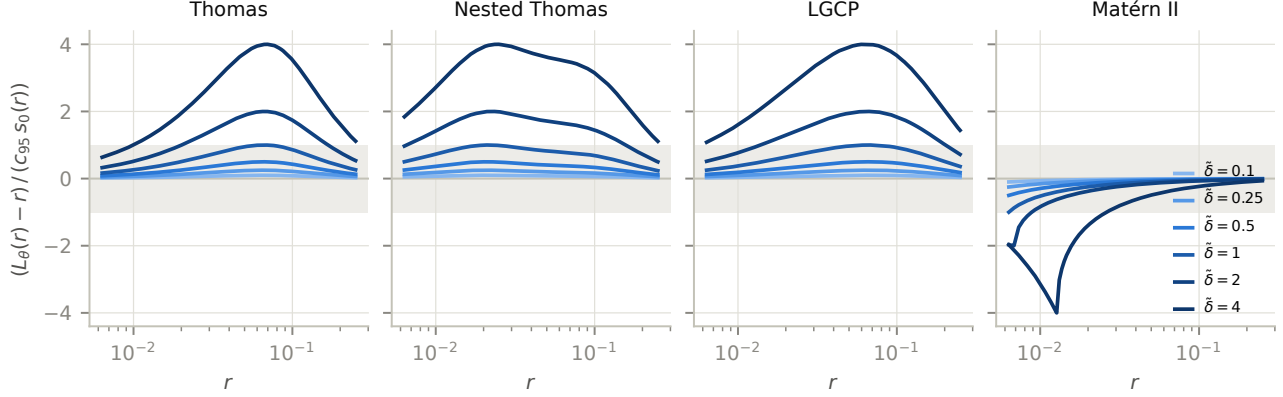


Figure 2: Closed-form $L_\theta(r) - r$ in units of the critical envelope, same θ as Fig. 1, on $R(400)$. Grey band: rejection boundary. Peak height = $\tilde{\delta}$.

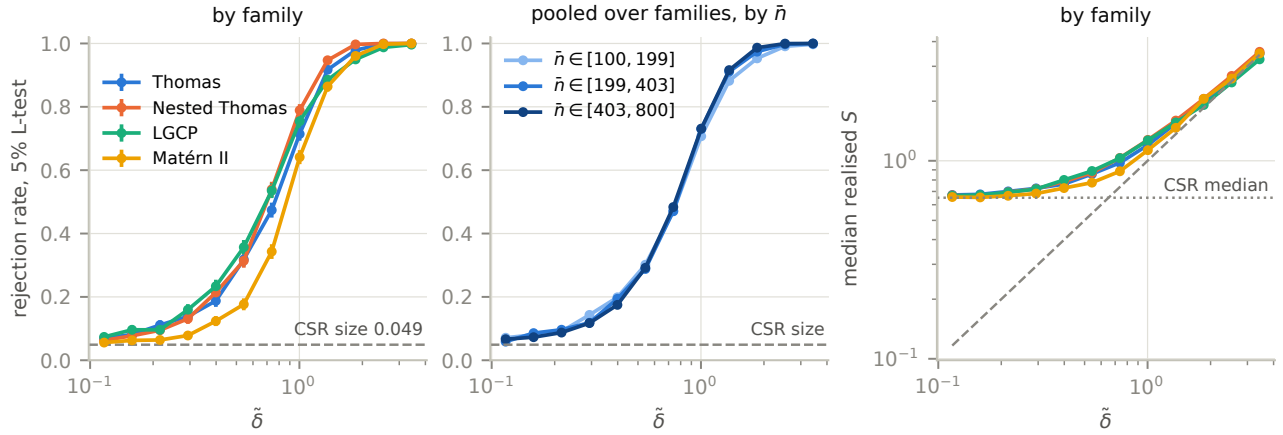


Figure 3: The 5% L-test of 01 on all 100 000 simulated patterns. Left: rejection rate by family, 12 log-bins of $\tilde{\delta}$, 95% intervals. Middle: pooled over families, by $\bar{n}$ tercile. Right: median realised S (dashed: $S = \tilde{\delta}$; dotted: CSR median).

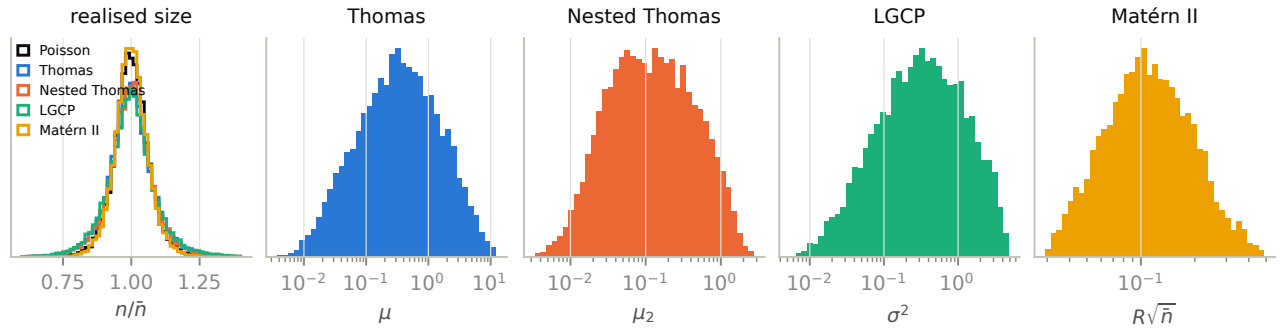


Figure 4: Left: realised $n/\bar{n}$. Right: induced distribution of the solved amplitude (Matérn: $\tau = R\sqrt{\bar{n}}$).

What the figures show

- **Convergence to Poisson** (Figs. 1, 2): along each row only the amplitude moves; curves shrink uniformly into the null band; at $\tilde{\delta} \leq 0.25$ patterns are visually CSR
- **Scale structure survives the amplitude:** nested keeps its two-step shoulder at every $\tilde{\delta}$; Matérn's deficit sits just above $r_{\min}(\bar{n})$
- **Same regime, same difficulty across families, for the L test** (Fig. 3, left; Matérn is much easier for a nearest-neighbour test, 01):

$\tilde{\delta}$ bin	(0.1, 0.2]	(0.2, 0.4]	(0.4, 0.8]	(0.8, 1.25]	(1.25, 2]	(2, 4]
Thomas	0.078	0.136	0.339	0.712	0.952	0.999
Nested	0.072	0.135	0.366	0.779	0.974	1.000
LGCP	0.085	0.152	0.386	0.746	0.923	0.988
Matérn II	0.059	0.084	0.214	0.639	0.914	0.997

- $\tilde{\delta}$ **absorbs** $\bar{n}$ (middle): terciles [100, 199], [199, 403], [403, 800] agree within 0.03 in every bin
- **Floor and ceiling:** below $\tilde{\delta} \approx 0.25$ rejection ≤ 0.15 and median $S = \text{CSR median}$ (0.65); above $\tilde{\delta} \approx 1.25$ rejection ≥ 0.91 and median $S \approx \tilde{\delta}$
- **Correction to 01:** at $\tilde{\delta} \approx 1$ rejection is 0.64–0.78, not $\approx 50\%$: pattern noise adds to the sup, so S tends to exceed $\tilde{\delta}$
- **Replicates:** the two patterns of a θ agree on rejection 77–82% of the time; correlation of $\log S$ 0.71–0.82

Coverage

$\tilde{\delta}$ **coverage.** Cells: 6 $\log\tilde{\delta}$ bins $\times$ 3 $\log\bar{n}$ bins, expected 556 θ each.

Family	realised $\tilde{\delta}$	$\log \tilde{\delta}$ uniform (KS p)	$\bar{n} \perp \tilde{\delta}$ (χ^2 p)	θ per cell
Thomas	0.1001–3.9998	0.76	0.81	530–581
Nested	0.1000–3.9969	0.98	0.54	519–587
LGCP	0.1000–3.9984	0.78	0.44	524–589
Matérn II	0.1001–3.9939	0.60	0.43	514–601

Parameter coverage at each $\tilde{\delta}$. At fixed $(\bar{n}, \tilde{\delta})$ the allowed parameters are the shapes passing rules 1–2 at $\bar{n}$; the amplitude is then determined.

- **By construction:** shape drawn after $\tilde{\delta}$, from its own uniforms; replay of every θ with $\tilde{\delta} < 0.2$ or $\tilde{\delta} > 3$: 0 shapes rejected by the amplitude solve (only nested geometric rejections) $\Rightarrow$ at every $\tilde{\delta} \leq 4$ the shape distribution is the full rule-1/2 prior, independent of $\tilde{\delta}$
- **Range:** normalised position $u = \log(x/x_{lo})/\log(x_{hi}/x_{lo})$ in the box at $\bar{n}$ spans 0.000–1.000 in every $\tilde{\delta}$ bin (Thomas σ , LGCP s , nested μ_1); nested σ_2, ρ reach 0.97–0.99 (top corner of the (σ_2, ρ) region excluded by the outer-scale rule)
- **Uniformity:** Thomas, LGCP uniform in u within each $\tilde{\delta}$ bin (KS p 0.08–0.94); u -quintile $\perp \tilde{\delta}$ bin (χ^2 p 0.10–0.81, all shape coordinates)
- **Nested:** vs a direct rejection sample of the allowed region at the same $\bar{n}$: pooled KS p 0.53–0.93; 3 of 18 per-bin tests < 0.05 (multiple testing; excluded as bias by the replay)
- **Amplitude range at fixed $\tilde{\delta}$:** wide, because the amplitude compensates for $\bar{n}$ and shape (e.g. larger $\bar{n}$ needs a weaker structure for the same $\tilde{\delta}$). Realised ($\tilde{\delta}$ within $\pm 10\%$) vs theory (grid over $\bar{n}$ and allowed shapes):

$\tilde{\delta}$	Thomas μ		LGCP σ^2		Matérn R	
	realised	theory	realised	theory	realised	theory
0.1	0.0037–0.32	0.0033–0.35	0.0055–0.28	0.0049–0.27	0.0010–0.0082	0.0010–0.0079
0.5	0.019–1.5	0.018–1.7	0.027–1.2	0.025–1.2	0.0022–0.017	0.0022–0.017
1	0.042–3.6	0.040–3.5	0.047–2.1	0.049–2.1	0.0029–0.022	0.0029–0.022
4	0.31–12	0.25–14	0.22–4.8	0.19–5.0	0.0059–0.047	0.0068–0.051

Theory grid coarse, so realised can exceed it

- **Density:** one-shape families ≥ 297 θ per ($\tilde{\delta}$ bin, shape quintile); nested ≥ 62 per ($\tilde{\delta}$ bin, marginal quintile), but jointly $\approx 10^4/(6 \cdot 5^3) \approx 13$ per cell

4 Statistical notes

Interpretation

- Regimes defined by one classical yardstick (L); visually, Matérn inhibition at $\tilde{\delta} \leq 4$ is hard to see by eye although the test sees it
 - $\bar{n} = 400$, $\tilde{\delta} = 4$: $R = 0.0127$ (0.25 spacings), $\pi R^2 \bar{n} = 0.20$; 25 fresh patterns all visually near-CSR, all with $S \in [3.76, 4.48]$; minimum pair distance $\geq R$ (sampler correct)
 - Removes only pairs closer than a quarter spacing; visibly regular patterns need fill ≈ 0.95 , i.e. $\tilde{\delta} \approx 8.6$ at $\bar{n} = 400$ (outside the common range)
- Matérn lags in $(0.2, 1.25]$: for $\tilde{\delta} \lesssim 1$ its hard core lies below $r_{\min}(\bar{n})$, outside $R(\bar{n})$, so $\tilde{\delta}$ and the test see only the deficit's tail (01). A nearest-neighbour test detects the same patterns with power 0.63–1.00 $\Rightarrow$ equal $\tilde{\delta}$ means equal difficulty for the L test only
- Shape distribution at every $\tilde{\delta}$ is the full rule-1/2 prior (Coverage): no shape rejected by the amplitude solve; nested redraws are geometric (the ρ – σ_2 coupling)
- Allowed shape box depends on $\bar{n}$ (lower bound $\propto r_{\min}(\bar{n})$): in absolute units the smallest scales occur only at large $\bar{n}$; in normalised u , shape $\perp \bar{n}$
- Matérn II has no shape: $(\bar{n}, \tilde{\delta})$ determines $R \Rightarrow$ regime-conditional inference error for Matérn conditions on almost the whole parameter (within a $\tilde{\delta}$ bin R still varies $\approx 8\times$ via $\bar{n}$ and bin width); not comparable one-to-one with families that keep shape variation
- Rule 3 inactive in this range (max realised CV $0.34 < 0.45$): it matters only if $\tilde{\delta}$ is ever raised
- Induced amplitude marginals (Fig. 4) are the effective inference prior: learned estimators shrink toward them

Scope and limits

- $\tilde{\delta}$ at $\bar{n}$; realised $n/\bar{n}$ 5–95%: 0.89–1.11 (Poisson, Matérn), 0.85–1.15 (clustered)
- Gallery and curves at one shape per family and $\bar{n} = 400$: illustrative, not exhaustive; the power table uses the whole pool
- LGCP: grid process, bias bounded at $0.1 s_0$; exact circulant embedding on a $2\times$ torus
- Thomas/nested buffer truncation $e^{-12.5}$

Open

- δ_{lo} : the classical test is at its floor below ≈ 0.25 ; decide with learned methods' curves
- Carving train/validation/test from the pool (by θ , both replicates together)
- Matérn below $\mu_R \approx 1$ (hard for every test) is not sampled: the $\tilde{\delta}$ floor 0.1 corresponds to $\mu_R \approx 1$
- Nested joint shape coverage thin (≈ 13 θ per $\tilde{\delta}$ bin $\times 5^3$ cell): enough for averaging over shape, not for joint shape analysis; more θ can be appended
- Shard files (data/simulation/shards/, 626 MB) redundant after merge